Aarav Kalkar

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Technical Skills

Languages: Python, SQL, R, Bash, PySpark
Data Science & ML: scikit-learn, XGBoost, LightGBM, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Survival Analysis, NLP, Information Retrieval
Experimentation & Analytics: Hypothesis Testing, A/B Testing, Regression Analysis, Time Series Analysis
Data Engineering: Pandas, NumPy, DuckDB, SQLAlchemy, BigQuery, REST APIs, FastAPI, Git, Vector DBs
MLOps & Reproducibility: MLflow, Docker, AWS, LangGraph, LangChain
Visualization & Applications: Tableau, Power BI, D3.js, Streamlit, React, GeoPandas, ArcGIS

Education

University of Minnesota, Twin Cities Minneapolis, MN, USA — Sep 2024–May 2026
Master of Science in Data Science — GPA: 3.75/4.0

SRM Institute of Science and Technology Chennai, India — Sep 2020–June 2024
Bachelor of Technology in Computer Science and Engineering — GPA: 9.01/10

Work Experience

Graduate Research Assistant – Applied Data Science — Sep 2024 – June 2026
University of Minnesota, Twin Cities – Minneapolis, MN

Predictive Education Analytics (Institute of Health Informatics)

- Engineered Python pipelines in a HIPAA-compliant environment to consolidate 10+ years of heterogeneous K-12 district data (125k+ records) into unified longitudinal datasets for cross-year modeling and stakeholder reporting.
- Trained XGBoost and Random Forest models to predict chronic absenteeism risk and GPA outcomes, applying propensity score matching to generate cohort-level risk indicators for longitudinal program evaluation.

Healthcare Risk Modeling (Department of Psychiatry)

- Developed XGBoost classification and survival analysis models across 100+ ICD diagnoses (n=3000), using RandomizedSearchCV to achieve 0.70 PR-AUC in rare-outcome neurodevelopmental risk prediction.
- Engineered an interactive D3.js/ArcGIS pro geospatial dashboard integrating patient data with 40+ socioeconomic overlays, enabling stakeholders to analyze structural risk patterns for research reporting.

Data Engineering & Longitudinal Survey Analytics (College of Design)

- Built Python and SQL pipelines to integrate and standardize fragmented schemas across 1.5M+ museum records, creating analysis-ready datasets for downstream modeling and D3.js visualization.
- Applied ordinal, logistic, and multinomial regression to 6 years of longitudinal survey and collections data, quantifying engagement patterns, donation likelihood, and temporal shifts for cross-year reporting.

Projects

RetentionIQ: Churn Decision System | *XGBoost, SHAP, React, scikit-learn* — Dec 2025

- Built a churn decision system across Kaggle and Online Retail II data, converting 821K transactions into 37,976 customer-month snapshots with temporal splitting and right-censoring.
- Trained XGBoost, Random Forest, and Logistic Regression against RFM baselines, achieving 85.7% Precision@10% and 5.1x lift with SHAP explanations and bootstrap CIs.

Spatio-Temporal Taxi Demand Forecasting | *PySpark, XGBoost, DuckDB, React, D3* — Jan 2026

- Engineered a PySpark Bronze/Silver/Gold pipeline over 47.2M NYC taxi trips, retaining 45.9M cleaned rows for grid-cell-by-hour demand forecasting.
- Benchmarked XGBoost, LightGBM, Random Forest, CatBoost, LSTM, and seasonal baselines; selected XGBoost with 0.0974 test wMAPE and hotspot/repositioning alerts.

RAG Evidence Lab: Retrieval Evaluation Workbench | *RAG, FAISS, BM25, Flask, React, D3* — Apr 2026

- Built a full-stack RAG evaluation workbench comparing five retrieval methods: Naive Vector, Hybrid, GraphRAG, Agentic Multi-hop, and Corrective RAG.
- Benchmarked 100 HotpotQA queries across 500 method-query evaluations, with Hybrid RAG achieving 0.7723 Recall@10, 0.99 Hit@10, and 0.8510 MRR.